

Some notes on GSAI-like plans

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1 Introduction

This is a discussion of some aspects of the GSAI plan (<https://arxiv.org/pdf/2405.06624>).

2 Some thoughts on what to keep in mind when evaluating plans for creating safe AGI

- The name of the game is to make a system which is capable (lets one do something transformatively useful) and safe, not just to make a system which is safe. Whenever we are considering an AI design, we should try to figure out if there's a decent chance that the proposed system would be more safe than an equally capable system which would be built by default.¹ Concretely, we could compare the safety of a proposed design to whatever system at the same level of capabilities we think OpenAI/Anthropic/DeepMind would build by default. Ideally, to pursue a design, we should have good reason to think that it is safer than such a default system at the same capability level. Minimally, we should believe that there is a 'significant chunk of worlds' in which the proposed design is safer than a default system at the same capability level.²
- The above is naive — for certain kinds of systems, we shouldn't think of capability as monolithic. The corrected question is: whenever we have a plan for building AI to solve some particular problem, we should compare it to some default plan for solving that problem, and ask which is more likely to provide a solution. (Actually, for certain kinds of deployments, we

¹Equivalently, we should try to figure out if there's a decent chance that the proposed system would be more capable than an equally safe system which would be built by default.

²It can make sense to pursue a design even if we think it has a smaller probability of turning out to be safe than some default option at the same capability level, as long as we think there is a good fraction of worlds in which the proposed design would later turn out to be safer than the default option — for instance, because we are currently uncertain about the value of some hyperparameter which critically affects the potential of the proposed design but not of the default design. In our case, for example, the feasibility of turning a meaningful chunk of scientific inquiry into narrow optimization might be such a hyperparameter.

might have significant probability on the deployment causing doom even up to the time of deployment, in which case we should have three outcomes in mind: [the system solves the problem] > [the system does nothing] > [the system causes doom]. This makes the analysis more complicated.)

- Suppose we have made a good-faith attempt to explore the subspace of plans of a certain kind in a fair bit of detail, and it looks like any plan in this subspace which we can easily hold in our head, or which we have already analyzed in detail, is unlikely to work. However, suppose there are versions of the plan which we have not examined in detail. In particular, suppose that these versions are more complicated than the versions we've already analyzed, with patches applied to address issues pointed out previously. In such cases, we should track some meta-estimate of the promisingness of future plans of the same kind, and abandon trying to come up with a good version of such a plan after enough evaluation. It's of course hard to say anything helpful about when that should happen without getting into the details.
- The above isn't to say that we need to track all of these factors super explicitly — while calculating things would be nice, it seems quite difficult. Even doing meaningful fermi estimation for whether some proposed change to a design leads to a better capability/safety tradeoff might be too difficult. I'm content with us instead just vaguely trying to keep the above in mind while evaluating plans.

3 General-domain plans

- One baseline plan here is to train a massive world-model, tell it the current situation (probably in natural language, with some learned translation which remains to be specified) — for example, that it looks like Safe Superintelligence Inc. has begun to paperclipify the Middle East (and various other news) — and seek an action such that the world model says that the probability the future will be good conditional on that action being taken is high. (Such an action might be found using some optimization on top of the world model — i.e., it wouldn't just be a primitive call to the world-model). Or maybe instead ask for an action such that the world model says that conditional on the future being good, the probability of that action is high.
- If we propose some meaningful split between the agent and the guardrail, then we should justify why that split would make the overall system safer at the same level of capability — this currently remains unclear to me.
- I have a hard time imagining how one could scale up the guardrail such that it keeps getting an increasingly better sense of what harm means to humans — we don't know how to get a very intelligent system to track human values (this seems almost alignment-complete).

- There are ways to scale up the guardrail and make the entire system safer which do not require the guardrail to get an increasingly better sense of human values, but I think the ones I can see plausibly just make the system paralyzed. For example, we could imagine an increasingly capable guardrail being tasked with providing arguments that a proposed action/plan is flawed, rejecting if a compelling argument is provided. We could scale up this guardrail while keeping the desired compellingness of the argument fixed. This will lead to more actions being rejected, so we get a less capable and more safe system. It's very unclear to me if this is actually an improvement compared to some default system at that same capability level.
- This relates to: it seems to me that there is no reasonable way to be infrabayesian, which seems central to at least some versions of the plan (<https://www.alignmentforum.org/posts/pKSmEkSQJsCSTK6nH/an-open-agency-architecture-for-commentId=jkk8xYoQiBTnjZEKj>). I discuss this more in Appendix A.
- This also relates to: inaction is not safe.
- It is hard to keep the agent from optimizing against the guardrail while keeping the system competent.
- To me, the most critical issue in all of this is that it seems unlikely that the guardrail will by default get a sufficiently good sense of human ethics to evaluate plans well. I'd like to have some concrete proposal to analyze here.

4 Narrow-domain plans

- The safety specification could just be merged with the task specification. It's really about solving narrow optimization problems. The safety specification makes it sound like we're doing some special thing beyond narrow optimization, but I do not think we are.
- I don't have a conclusive argument for the claim that the natural class of problems is not those which can be solved with a Davidad-style system, but narrow optimization problems. The best I can do is the following.
 - For most examples, the split into a safety and task specification feels shoehorned. It feels like the problem could be more naturally stated as just having a task specification.
 - It feels like similar problems could be stated where there wouldn't be anything reasonable to put in the safety specification at all; it feels very natural to consider these in the same class of problems.
 - For almost all examples, the safety specifications have little to do with safety in any broad sense, in the sense of alignment, just about

avoiding particular narrow failures. For example, in the safe operating system, the specification is about computer security, which captures a very small fraction of what we'd want for the operating system to do good stuff in the world in general.

- How much of weight in guaranteeing safety of the impact of the potentially misaligned optimizer is pulled by the task being narrow, and how much is pulled by the formal safety specification? It seems to me that in most cases almost all the weight is pulled by the task being narrow.
- The obvious issue is that it is hard to find a plan for using narrow optimizations to act transformatively. It is hard to get narrow optimizations to 'do a lot' — in particular, to 'do a lot compared to humanity'.
- "If you tell me precisely what it is a machine cannot do, then I can always make a machine which will do just that." – John von Neumann
- It is hard to make scientific problems precise. Philosophy is all about making things precise, and philosophy is not easy. Science also does a lot of making questions precise.
- Any scientific work generated will not (by default) be integrated into our understanding. (Note how it is unclear whether even math-theorem-prover-ASI could be used further our mathematical understanding that meaningfully. We might need to make its proofs legible to us, make sense of new objects it defines, understand the informal patterns it seems to be employing, etc.. If it just outputs proof strings for particular problems, it might not even make it so that humans are no longer needed to push the frontier of mathematics — after all, to push the frontier, one needs to keep asking interesting questions. In some sense, the questions are there only to indicate where there's interesting mathematical territory we haven't yet understood.)

5 Some things about/[relating to] the GSAI plan that I consider hopeful

- I think there's some hope in **trying to find a pivotal act plan which can be reduced to narrow optimizations, and then using AI to do these narrow optimizations**. I don't know of any particular plan like this that seems remotely plausible, though.³ I think it could be worth

³I guess the obvious high-level issue is that to pull off a pivotal act without having broad popular support, one sort of needs to 'win a battle against most of humanity', for which one sort of (at least in some particular context) needs to be 'a more powerful general optimizer than humanity'. Still, there might be some reasonable way to do this without an artificial general intelligence being involved, with humans directing all real-world optimization and using the outputs of some narrow optimizers.

looking into doing mind uploading this way. (I also have some hope about pulling off mind uploading with usual human researchers, though.)⁴

- **Trying to find a way to decrease x-risk from AI using a pretty serious governance solution together with narrow optimizations** also seems pretty reasonable. I think I currently have a hard time imagining such things, but I have also thought about these less than I have thought about pivotal acts which require less buy-in from humanity.⁵
- I think there's some hope in **using narrow AI to radically speed up humans doing science, philosophy, and engineering**. For example, perhaps one could develop an AI-augmented note-taking system which points at places where your thinking seems most critically confused, or points out to you when you've had analogous thoughts in different contexts, or helps you remember other notes you've written that relate to the note you're currently writing, or suggests relevant academic literature. Of course, none of this sounds that revolutionary. One major issue here is that one probably needs to get a pretty large advantage over others to make this useful w.r.t. x-risk, and this is difficult given that lots of people have tried to do things like this for a long time. (If there were a way to broadly distribute such a thing to alignment researchers and not capabilities researchers, even modest gains could make this worthwhile, but such secrecy seems incredibly hard to pull off.) Another issue is that maybe we spend a significant fraction of our thinking compute on each of some number of 'sorta distinct processes', in which case we'd need to improve all of them by a lot to get a major improvement (otherwise the few we fail to speed up would still take up too much time for the entire process to be, say, 10 times faster).

One major issue with any of these plans, except maybe the second depending on the version, is that might require one group to develop a massive edge over other groups. Another thing that I dislike about these plans is that they don't really directly tackle the problem of designing a safe ASI, instead just involving deferring to others, creating/bettering others to defer to.⁶

⁴The hope would be that one could then have sped-up uploaded researchers figure out how to create an aligned superintelligence or devise/implement some other plan toward a safe world, potentially starting from improving their own thinking.

⁵I can sort of imagine 'cynical ways' to do that — imagine something like the following but more sophisticated: telling a government that one can build a safe superintelligence in such-and-such a way and they should mandate that anyone this is the only allowed way to build a superintelligence, while the way really only lets one construct a system for solving linear programming problems or something. But I think one shouldn't try to sell the government a rock with 'safe superintelligence' written on it (for one, because other people will likely see through such an attempt and be outraged).

⁶I have some probability on this last thing being sort of necessary though, so I'm not too negative about it.

A Against maximin decision rules

remark: This is copied from a LW question I started writing but didn't finish yet.

I take some people to view the following line of reasoning in favor of maximin favorably: * The real world is too complicated for a realistic system (inside it) to 'fully hold in its head', precluding e.g. some ideal form of bayesianism in which (e.g.) all possible 'world-processes' that might be generating one's stream of observations are held in one's head and updating corresponds to crossing out the processes which are in contradiction with what one has previously observed. It makes sense to try to make⁷ an intelligent system which has a partial model of the world, and, in the interest of safety, picks an action which makes the worst-case (expected) outcome as good as possible, where worst-case means worst-case over all specifications of the parts of the world-process 'outside the partial model of the world'.

One can also give a parallel line of thought where instead of or in addition to maintaining only a partial model of the world that could be extrapolated in various ways which are to be maximin-ed over when making decisions, one maintains only a partial model of goodness (or neg-harm) that could be extrapolated in various ways which are to be maximin-ed over when making decisions. This would be motivated by an analogous thought that while values might be too tricky to pin down, one can perhaps provide a correct partial specification (catch our true values in some convex set in some kind of value-space, say), with the minimax rule w.r.t. the remainder of the specification being motivated by an aspiration to act cautiously.

This seems wrong to me — I think holding a partial model over whose completions one is maximin-ing which is actually helpful for addressing the issue that it is hard to pin down a model correctly enough to just maximize expected utility yields either paralysis or being guided entirely by some comparison of ε 's which is hard to analyze.⁸

unfinished: give bit counting argument. for this to be useful, there needs to be a significant number of bits remaining in the specification, but then for any action, one can cook up a weird specification which makes it look very bad. one then compares some ε utilities to act in the infrabayesian case, and some $1 - \varepsilon$ probabilities of harm in the davidad case.

⁷or, to begin with, to contemplate as a serious candidate for an ideal version of a realistic intelligent aligned system

⁸I also find it suspect that the verdict of such a decision rule would be any easier to compute than that of naive bayesian expected utility maximization — that is, I don't see reason to think it wouldn't be equally impossible in any realistic case. Perhaps the point is that each decision rule is to be approximated instead, and it is for some reason easier to approximate the knightian one while getting a safely acting agent? I don't see why that would be the case. Anyway, I don't want to focus on this objection.

B What could we do with a narrow optimizer?

In this section, I'll be supposing that we have a math-ASI — i.e., a machine which can solve narrow optimization problems at a somewhat superhuman level (say at the level of a great human mathematician-scientist that has a lot of time to think), and describing some practical problems which we could plausibly use it to safely solve.

I'm aware many of the problems described below can be seen as instances/subsets of problems known to be very hard (for instance, NP-hard or plain uncomputable). Still, I think of this discussion as obviously much more relevant to the real world than discussion of what we could do in some arbitrarily weird hypothetical world. How come? Well, the problems below being embeddable in hard classes provides fairly little reason to think that it's infeasible to create an algorithm which solves most particular instances of interest efficiently — such an efficient algorithm need not solve all (or even most) instances in the hard class. And in fact, it seems plausible that when it becomes feasible to create a general-domain human-level AI, it will in particular be feasible to create a mathematical human-level intelligence, and creating a somewhat superhuman math AI is then also probably within reach. Furthermore, it seems plausible that we could safely use such a superhuman math AI, only having it solve certain mathematical optimization/search problems, only interacting with the world by receiving problem statements and outputting solutions which are used in limited ways. If we could find a way to use a black box solving formally specified optimization problems to move the world toward existential safety, then we'd have a plausible path toward a good future. In conclusion, which real-world things can be done given a somewhat superhuman math AI is of great practical relevance.

To make the exposition of possible optimization problems to unleash our math-ASI on simpler, I won't stop to assess whether each problem listed is actually reasonably close to the human range, and so could be expected to be solved by a math-ASI which is only somewhat superhuman (specifically, one upper bound on the capability of a math-ASI we could use is set by the point at which we would expect the math-ASI to take over the world by 'doing weird stuff'; another upper bound is set by practical considerations having to do with other kinds of AGIs being built at some point, making this effort irrelevant). So if my list below contained all the things one could do with a narrow AI, then a practical math-ASI doing these provides an upper bound on what could be done with a math-ASI. And in that same conditional, if these are insufficient for averting doom, then doom cannot be averted with a math-ASI.

B.1 A sampling of problems

Finding good algorithms for any formally specifiable problems. For example, you can ask the math-ASI for an algorithm that sorts lists given some particular space and time constraints. You can ask it to try to provide a $2^{0.99n}$ -time algorithm for SAT, or a polynomial-time algorithm for SAT. You could ask

it for a proof that

(relatedly) Finding efficient data structures. For example, you specify that you want to perform operations x , y , and z , and you ask for a specification of a data structure which makes a certain linear combination of the (asymptotic) times it takes to perform these operations minimal.

much more stuff missing here

B.2 How does this compare to being able to solve the above problems yourself?

- Sometimes you reduce a practical situation to some problem, you try to solve it, you realize it's not quite the right problem. This kind of thing seems impossible (absent some ways to interpret what's going on inside the math-ASI, anyway) if you just ask a system to solve the problem and see the answer it provides.
- Given this, perhaps we currently have some good (scientifically/technologically relevant) mathematical questions to ask, but then if we don't struggle on these ourselves, we might not be able to continue asking useful questions in the future (without returning to struggling on problems ourselves despite knowing the answer).
- I haven't investigated the relevant history carefully here, but it's my impression that technologies are often found when trying to build completely different technologies. This possibility to use subcomponents developed during an investigation for completely different purposes is diminished if the makeup of the investigation becomes opaque to us.

C On the feasibility of providing a formal language in which philosophical questions could be correctly asked and answered

This is my current draft for a paper I'm considering writing for this essay contest on automating philosophy: <https://www.lesswrong.com/posts/52ygLry5KCdvxY6zn/essay-competition-on-the-automation-of-wisdom-and-philosophy>.

C.1 Introduction

Safely getting meaningfully superhuman mathematical problem-solving out of an artificial intelligence seems feasible. A reason for this is that, since about 1925, we've possessed a way to clearly state mathematical problems and a precise notion of what it is to provide a solution to a particular mathematical problem. For example, the following is the twin prime conjecture written in the language of set theory (meta-remark: idk if this example is the best one or if an example is necessary. it will look fucked if we have to translate to base ZFC. perhaps an annotated version would make sense, ie where we eg point out which part just defines the naturals or whatever):

$$\forall n \in \mathbb{N}, \exists p > n...$$

A proof of the twin prime conjecture corresponds to a sequence of strings in this same language such that each string is either an axiom or is produced by the application of some inference rule to some strings which appear earlier than it in the sequence. Importantly, as it is decidable whether a given string is an axiom and decidable whether a given string can be produced by some inference rule from some strings from a given finite sequence of strings, it is also decidable whether a given sequence of strings constitutes a proof of a given statement. (Indeed, software exists for checking proofs written in the language of set theory: https://en.wikipedia.org/wiki/Mizar_system.) This makes it look plausible that, conditional on us figuring out how to train an artificial system which is better than humans at general reasoning, we could

finish the explanation why having a formal language plausibly lets us plausibly safely get superhuman mathematical work out of an artificial system. give more examples of such work: the problem of providing an algorithm to solve some well-defined problem, the problem of constructing an efficient data structure which supports some specific queries, the problem of providing a causal model which predicts certain data well, various other mathematical optimization problems

then say that it'd be great if we could do the same for philosophical questions — it would open up a path to safe ASI. say the point of this essay is to understand more clearly what doing this for philosophy would look like + whether it would be possible for philosophy + why it would be harder for philosophy than for mathematics

motivate this by saying that (1) mathematics and philosophy both sorta seem a priori, (2) historically, both used to be done informally, and people sorta set out to formalize them together (Frege), but then we succeeded at formalizing matheamtics but haven't yet succeeded at formalizing philosophy, (3) people xyz hope to do something like this

address the objection that well obviously philosophy is harder to formalize because philosophy just consists of the problems that we are still struggling to make precise, that contain terms we're still confused about, by saying that ok, even if that's precisely what philosophy is, we are asking instead [what it would look like] / [whether it is feasible] / [why it would be hard] to formalize the following particular example problems (or, more precisely, which common properties of these problems make them harder to formalize than mathematical problems): * list of examples *

meta-remark: actually, i think i'm a bit unhappy with the paragraph above. we could make the rest of the paper such that it makes sense, but i don't think it fits currently. the issue is that we decide to go with discussing the difficulties with constructing a language in which such questions could generally be formalized later, not with what it'd take to formalize some more particular questions. i've put a slightly modified version below that i think makes more sense given the current plan for the rest of the paper, but i've not erased the one above because i haven't thought about this change carefully yet

address the objection that well obviously philosophy is harder to formalize because philosophy just consists of the problems that we are still struggling to make precise, that contain terms we're still confused about, by saying that ok, we are instead asking [what it would look like] / [whether it is feasible] / [why it would be hard] to create a formal system in which the following example problems can be stated and answered: * list of examples * if the construction of a formal system subsuming in which these problems can be discussed would take them outside of philosophy in your view, then you should think of us not as trying to turn philosophy into something that could be carried out formally, but to replace it with something that could be carried out formally

C.2 Clarifying what we'd want from a formal philosophical system

might want to make the problem clearer, eg is this formalization thing something that happens after things fall out of philosophy by becoming sufficiently clear, or are we trying to formalize this process of making things precise itself, or are we doing some other thing?

let's say we're going with the ambitious goal of formalizing general philosophy here, not particular slices; re the list of questions above, we're not asking why it is that any particular question does not have a fo (meta-remark: I'm not sure this is the right choice of focus. it would also be reasonable to examine other options.)

distinction: note that when proving something in mathematics, most reasoning doesn't proceed like the eventual proof, and in fact it seems plausible that

most reasoning wouldn't faithfully fit in the formal systems known at present at all. consider: playing around with some examples to 'get a sense' for some particular kind of object, or thinking about which kinds of objects one would need to play around with to get a better sense for what makes a particular problem difficult, or thinking about the high-level similarities between two arguments to encapsulate the common thread in a lemma, or etc (meta: can give many more examples here — i think most mathematical reasoning cannot be formalized in existing systems). still, for math, this is fine for our purposes of getting mathematical work

now make this same distinction for philosophy: try to state the problem of having a language in which philosophical problems and arguments can be stated, and make it clear that this could be helpful even if captures only a small fraction of the reasoning that goes into producing philosophy

C.3 What are the obstacles to carrying out this formalization?

* it seems likely that a formal language in which philosophical arguments could be provided would imply a formal language in which any arguments could be provided, including eg arguments inside particular scientific theories, arguments that particular scientific theories are correct, arguments that a correct scientific theory would need to have such-and-such properties. so formalizing philosophy plausibly cannot be done without formalizing everything. the converse, that science (or just ordinary more descriptive reasoning) couldn't be formalized without philosophy, is also plausible. maybe it's clearer to ask for a formalization of everything * despite being sorta a priori, sufficiently tied to the world for probabilistic/uncertain/defeasible argumentation to probably have to show up in any satisfactory formalization. give examples. talk about how people are working on this * maybe to get a meaningful chunk of philosophy, one really needs to effectively have a language in which all the informal reasoning that shows up in mathematics but hasn't been formalized yet could be formalized, because these informal questions can be asked as philosophical questions (to do: give examples). so maybe the distinction between all reasoning which goes into some conclusion and just statements and proofs/arguments doesn't help us (doesn't imply that our formalization project is more feasible) after all * minimally, it seems likely that we will often need any compelling final argument to essentially deconfuse us about the question itself — it will sometimes need to restate a better version of the question. this is roughly because the deconfusion work cannot happen on the step where the problem is turned into a formal problem, because we don't want to offload this philosophical work to the system, not have to do it ourselves. so we need to be able to state confused questions in the formal language, and then provide arguments answering them; these arguments will probably have to involve stuff that looks like restating the problem, arguing about restatements, then solving it (insofar as anything remains to be solved at that point). in some sense, i think we are in extra trouble because 'a confused question cannot be translated — it only makes sense in the language in which

it was asked' (meta-remark: but this obviously needs to be made more precise)
* normativity: note that if we had a language in which the question "should i do x?" can be appropriately stated and pertinent arguments can be provided, then it must immediately compel us * more sophisticated point, with examples, on philosophical inquiry being purposive * insofar purposes exist already, they are stated in our current informal language — this ties philosophical reasoning to our current informal language

C.4 conclusions

seems hard to formalize, because might need to 'contain humanity' in the system setup or something, also blabla (just summarize previous section). relatedly, a system which formalizes would plausibly need to have properties xyz. all this plausibly applies also to any even slightly broad questions in science, though we haven't argued for this that much.